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Securely call Logic App Workflows with API Management

Andrew Wilson

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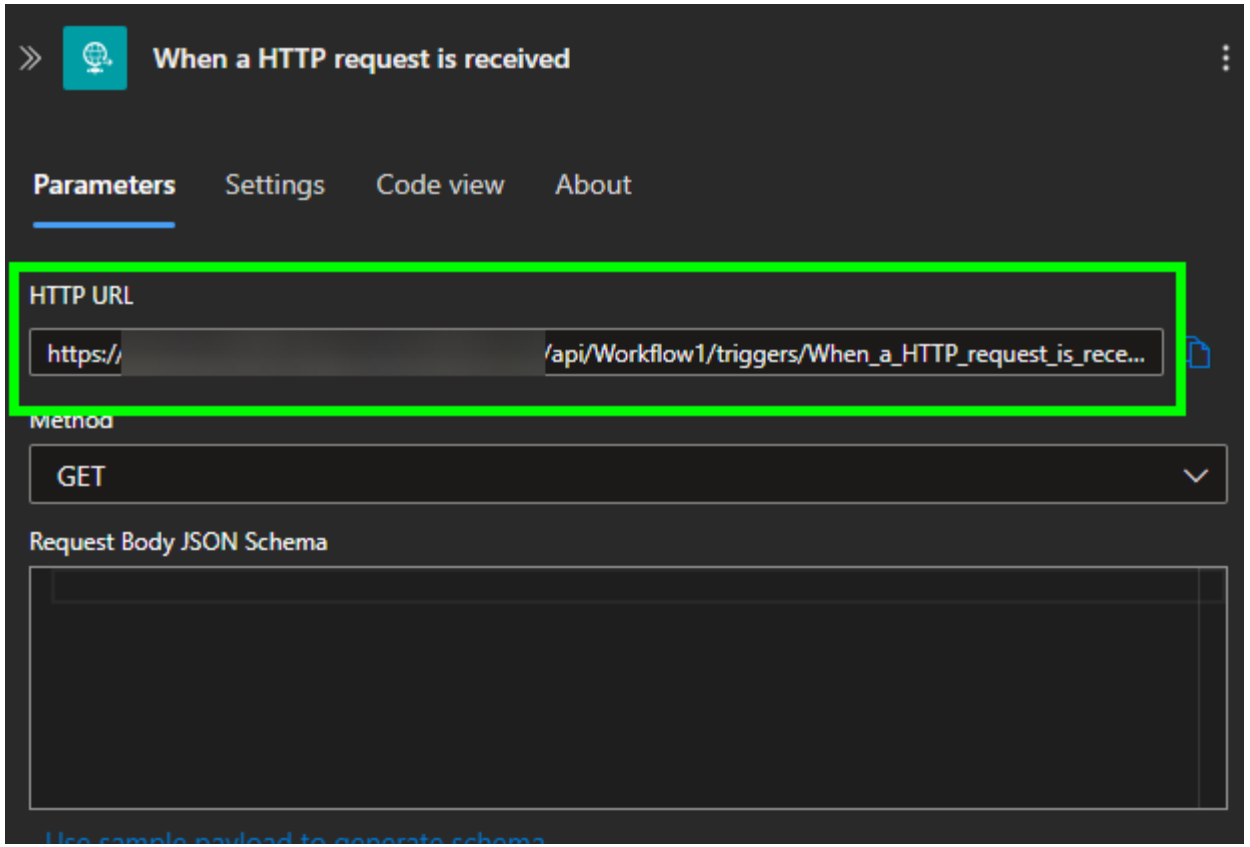

When a HTTP request is received
Parameters
Settings
Code view
About
HTTP URL
https:// /api/Workflow1/triggers/When_a_HTTP_request_is_rece...
Method
GET
Request Body JSON Schema

https://dddtestlogicapp.azurewebsites.net

/api/Workflow1/triggers/When_a_HTTP_request_is_received/invoke?
api-version=2022-05-01&

SAS query parameters:
sp=%2Ftriggers%2FWhen_a_HTTP_request_is_received%2Frun&
sv=1.0&
sig=w9JT9cne3w4sdfoXNx7r9VDJ3isdfsIKmvCu0V2hosdf

Backend

HTTP(s) endpoint

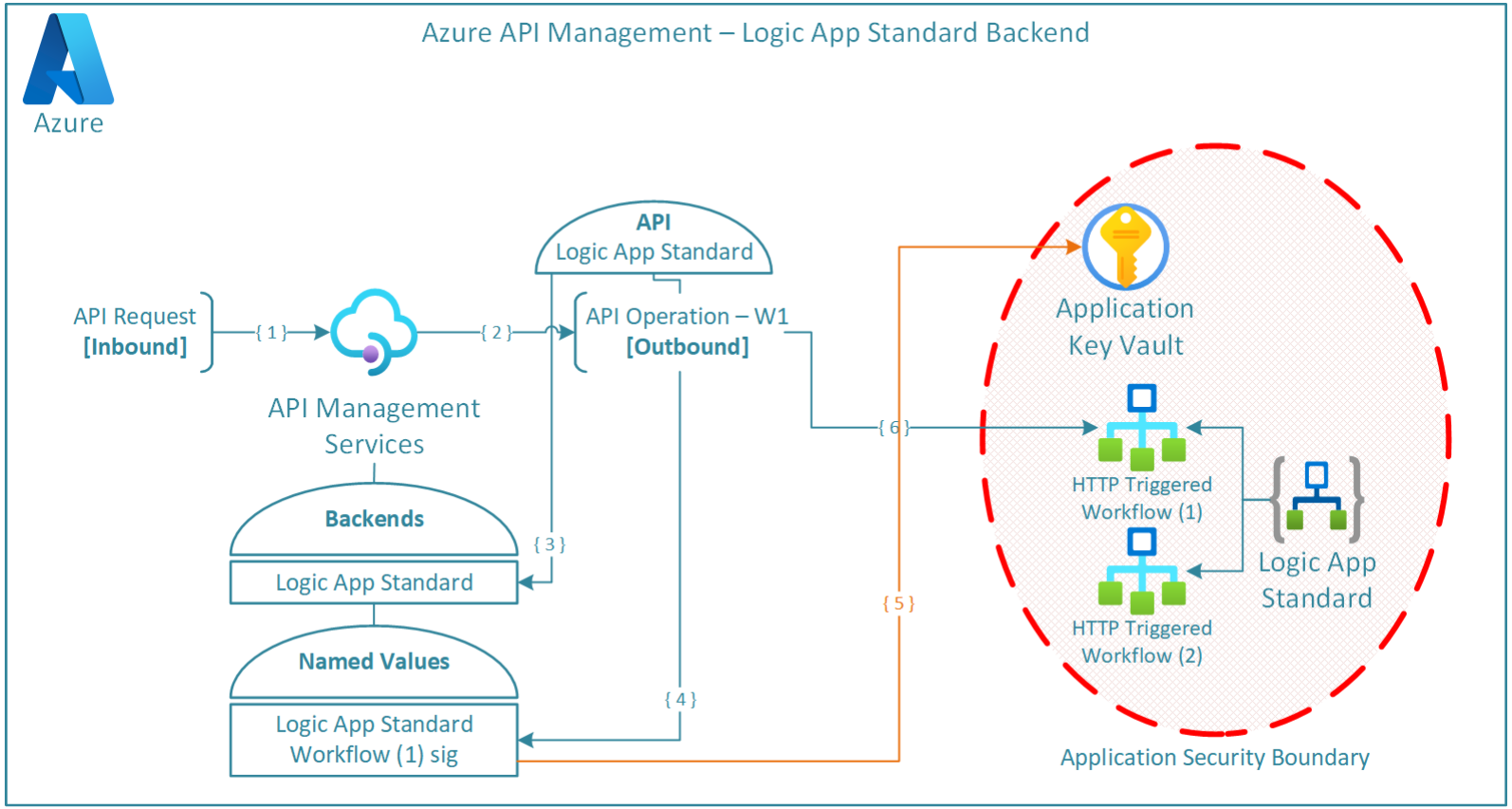
https://dddtestlogicapp.azurewebsites.net/api/Workflow1/triggers/When_a_HTTP_request_is_...

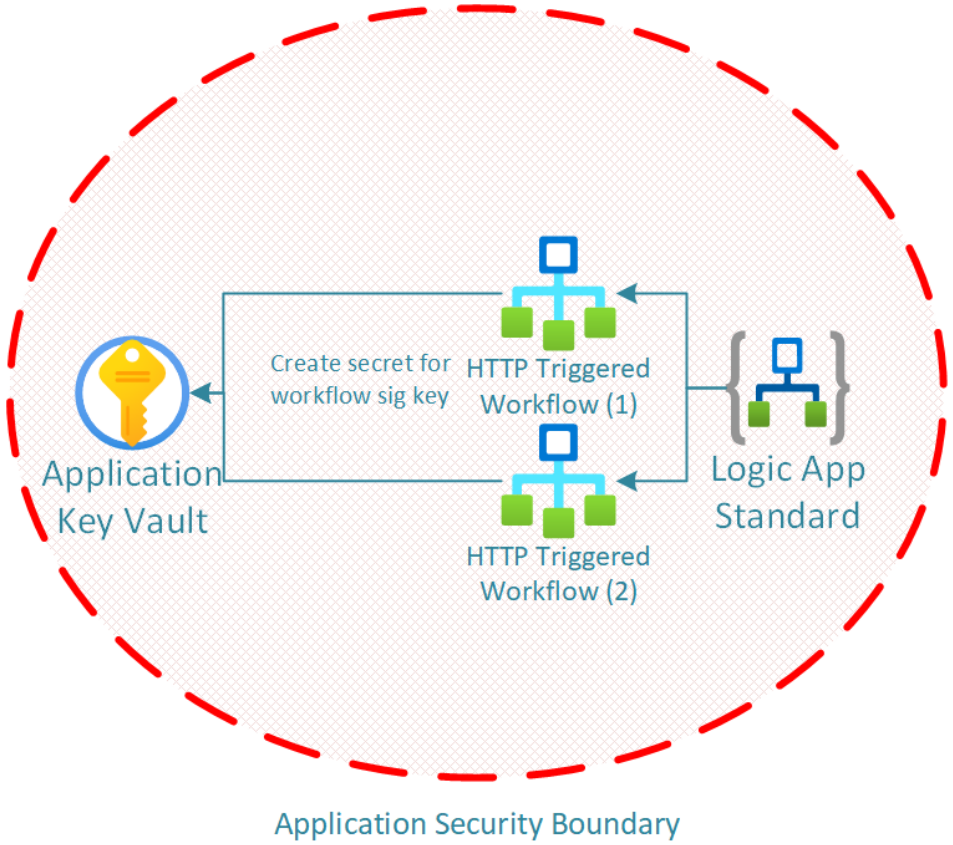
Policies

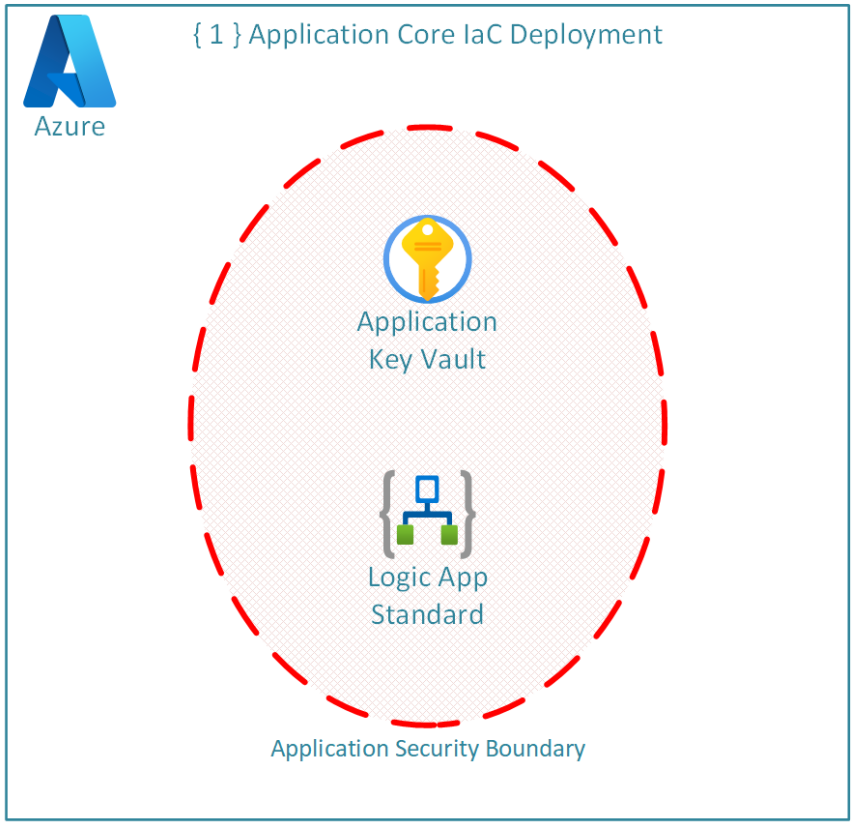
base

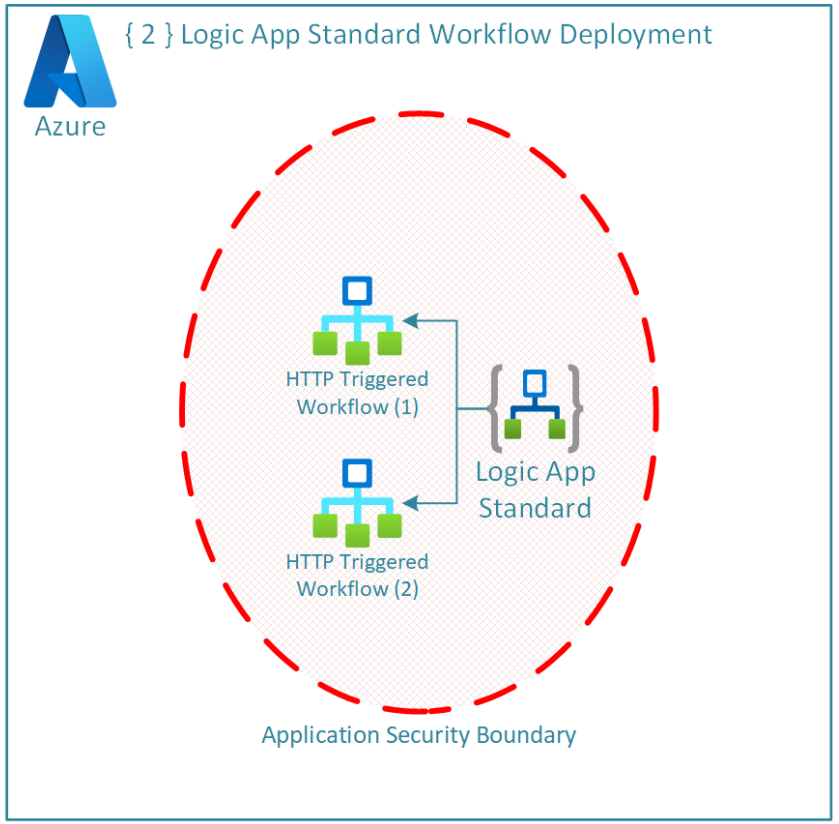
SAS Leak Mitigation

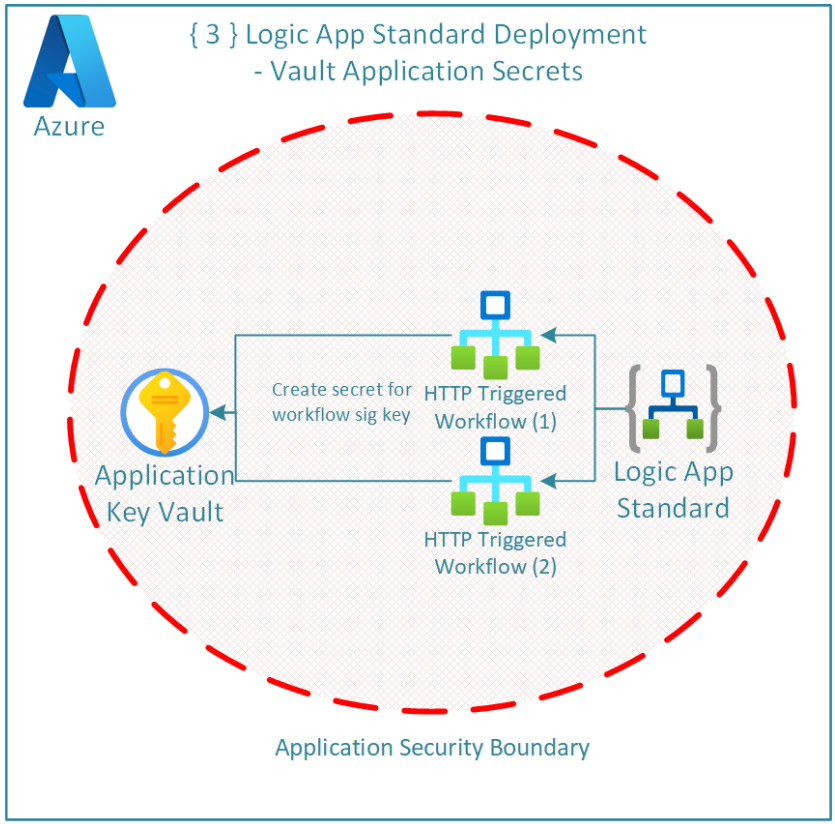
- Always use HTTPS
- Have a Revocation Plan
- Have a Rotation Plan

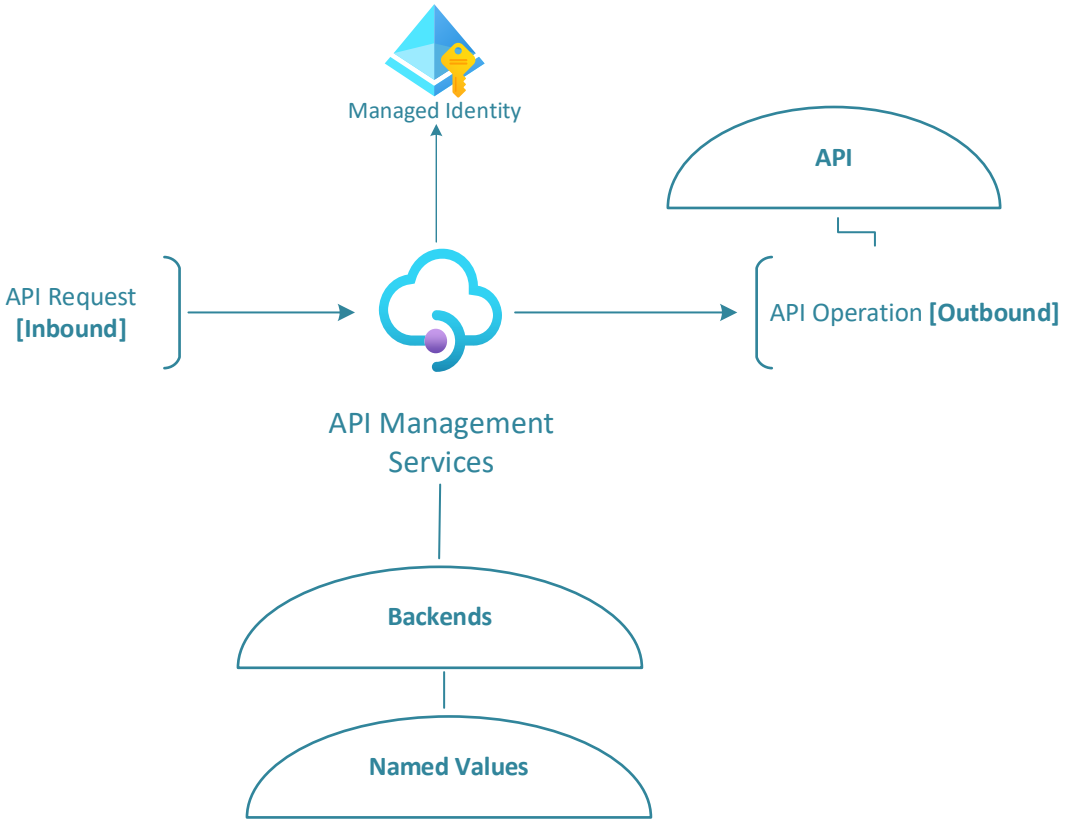

Azure API Management – Logic App Standard Backend
Azure
API Request
[Inbound]
{ 1 }
API Management
Services
{ 2 }
API
Logic App Standard
API Operation – W1
[Outbound]
Backends
Logic App Standard
{ 3 }
Named Values
Logic App Standard
Workflow (1) sig
{ 4 }
{ 5 }
{ 6 }
Application
Key Vault
HTTP Triggered
Workflow (1)
HTTP Triggered
Workflow (2)
Logic App
Standard
Application Security Boundary

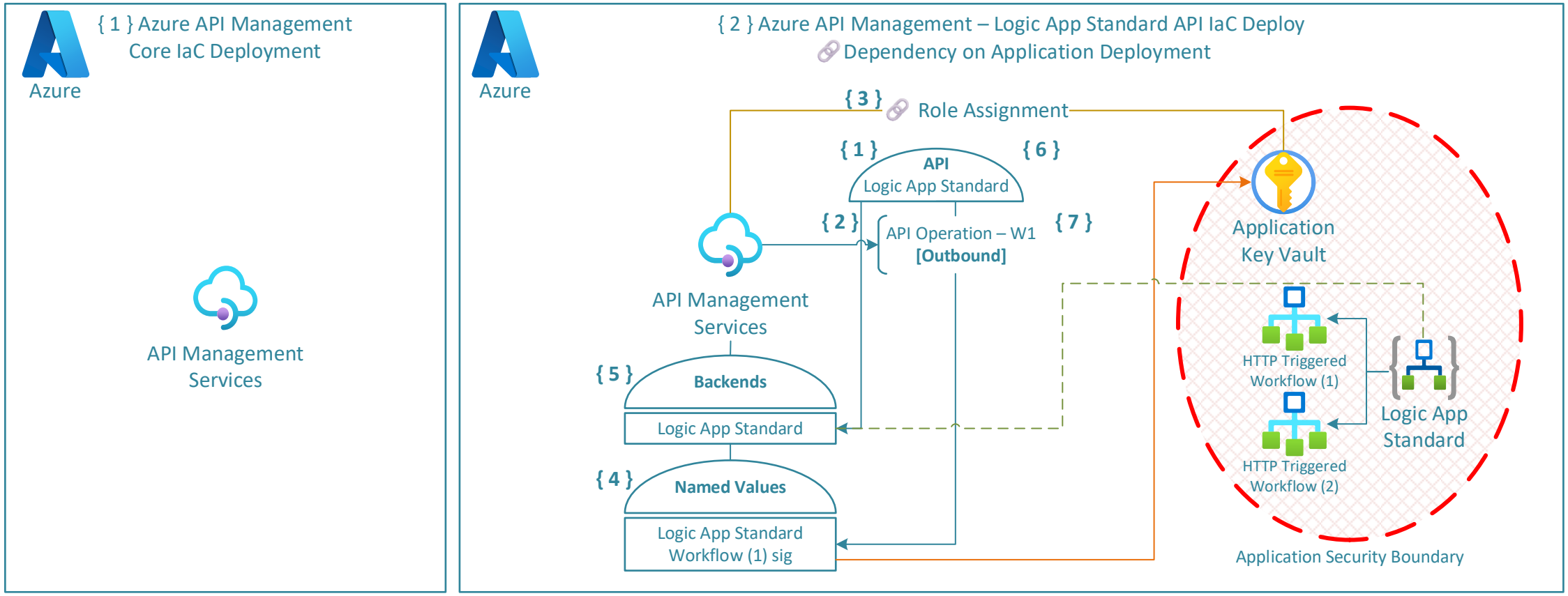

Create secret for workflow sig key
HTTP Triggered Workflow (1)
HTTP Triggered Workflow (2)
Application Key Vault
Logic App Standard
Application Security Boundary


{ 1 } Application Core IaC Deployment
Azure
Application
Key Vault
Logic App
Standard
Application Security Boundary


{ 2 } Logic App Standard Workflow Deployment
Azure
HTTP Triggered
Workflow (1)
HTTP Triggered
Workflow (2)
Logic App
Standard
Application Security Boundary


{ 3 } Logic App Standard Deployment
- Vault Application Secrets
Azure
Create secret for
workflow sig key
HTTP Triggered
Workflow (1)
HTTP Triggered
Workflow (2)
Application
Key Vault
Logic App
Standard
Application Security Boundary


Managed Identity
API
API Request
[Inbound]
API Operation [Outbound]
API Management
Services
Backends
Named Values


Azure
{ 1 } Azure API Management
Core IaC Deployment
API Management
Services
Azure
{ 2 } Azure API Management – Logic App Standard API IaC Deploy
Dependency on Application Deployment
{ 3 }
Role Assignment
{ 1 }
API
Logic App Standard
{ 6 }
{ 2 }
API Operation – W1
[Outbound]
{ 7 }
API Management
Services
{ 5 }
Backends
Logic App Standard
{ 4 }
Named Values
Logic App Standard
Workflow (1) sig
Application
Key Vault
HTTP Triggered
Workflow (1)
HTTP Triggered
Workflow (2)
Logic App
Standard
Application Security Boundary

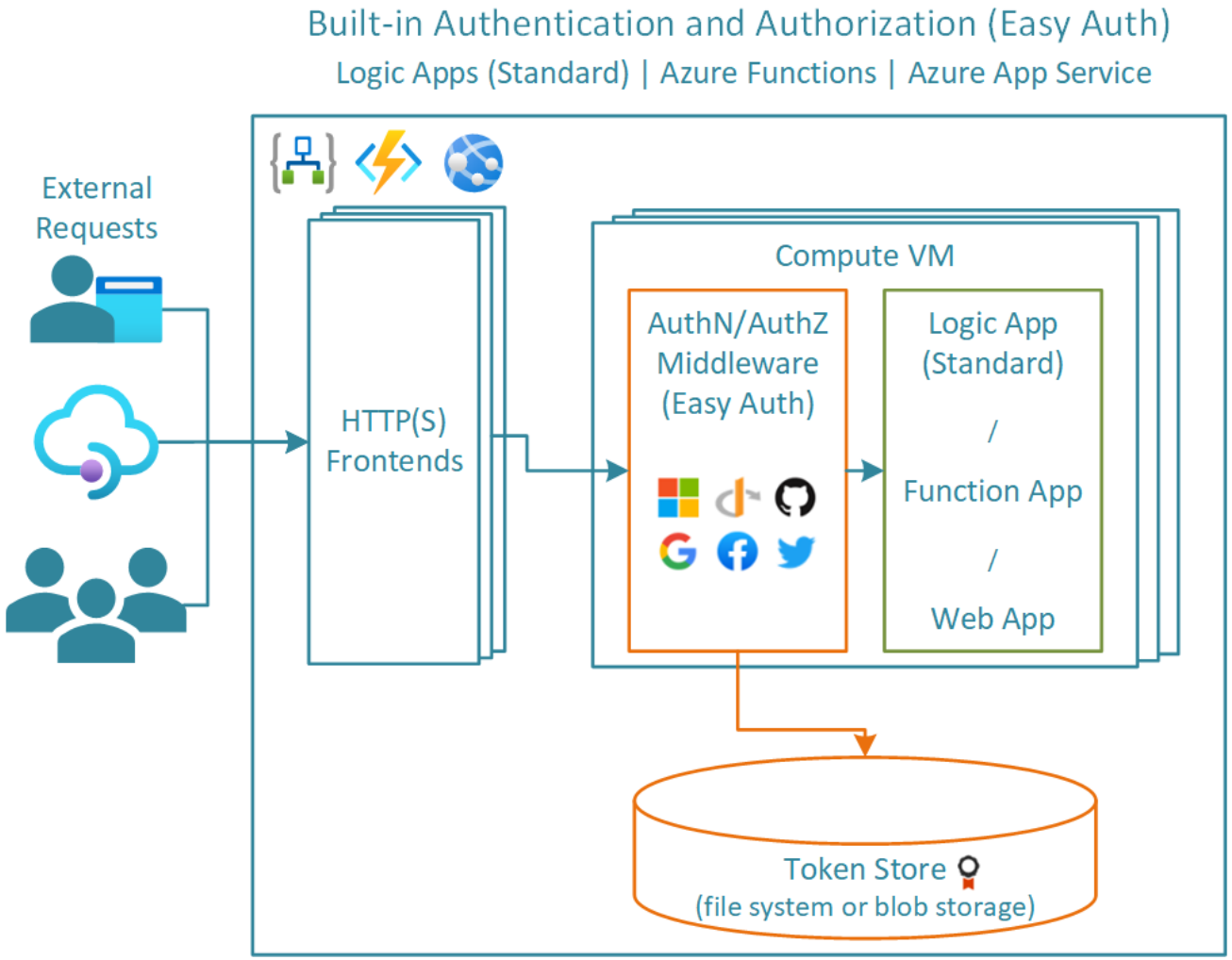
Logic App SAS Summary

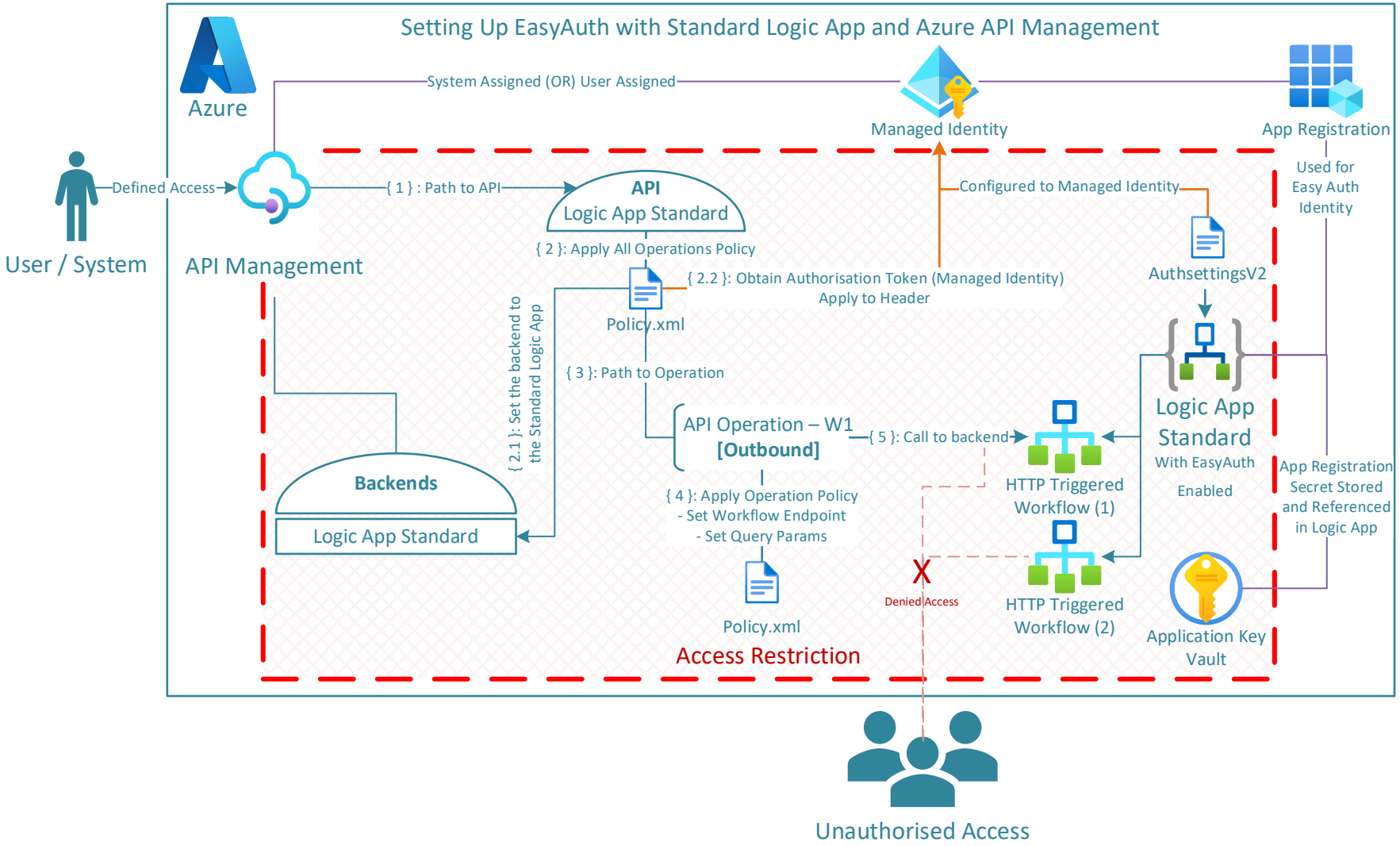
- Store and Access Securely
- Adhere to SAS key Best Practices

Is SAS the Best Way?

- Can we use Identity?

Easy Auth


Built-in Authentication and Authorization (Easy Auth)
Logic Apps (Standard) | Azure Functions | Azure App Service
External Requests
HTTP(S) Frontends
Compute VM
AuthN/AuthZ Middleware (Easy Auth)
Logic App (Standard)
/
Function App
/
Web App
Token Store
(file system or blob storage)


Setting Up EasyAuth with Standard Logic App and Azure API Management
Azure
System Assigned (OR) User Assigned
Managed Identity
App Registration
Used for Easy Auth Identity
User / System
Defined Access
API Management
{ 1 } : Path to API
API
Logic App Standard
Configured to Managed Identity
{ 2 }: Apply All Operations Policy
{ 2.2 }: Obtain Authorisation Token (Managed Identity)
Apply to Header
Policy.xml
AuthsettingsV2
{ 2.1 }: Set the backend to the Standard Logic App
{ 3 }: Path to Operation
API Operation – W1
[Outbound]
{ 5 }: Call to backend
Logic App Standard
With EasyAuth Enabled
Backends
Logic App Standard
{ 4 }: Apply Operation Policy
- Set Workflow Endpoint
- Set Query Params
HTTP Triggered Workflow (1)
App Registration Secret Stored and Referenced in Logic App
Policy.xml
Denied Access
HTTP Triggered Workflow (2)
Application Key Vault
Access Restriction
Unauthorised Access

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Thank you for your time

Any Questions?